

Session 7 – Exploring and Mapping Social Landscape of London (Basic Geostatistics and Thematic Mapping with QGIS and GeoDa)

Objectives: To develop skills of Geostatistics and Thematic Mapping through QGIS and GeoDa functionalities.

Task 1: Exploring Health Data across London Boroughs

Problem solving: Initially reveal spatial pattern of public health in London

Functionality: Basic Geostatistics in GeoDa

Data set: Health variables by London Borough

In GeoDa

Open a new project in GeoDa: File -> New, Input the Shape file of "london_life_polygon.shp"

Practice following statistical functionalities with health variables of 'LIFE_MALE' (male life expectancy), 'LIFE_FEMALE' (female life expectancy), 'P_SMOKE' (percentage of smoking population), 'P_BINGE' (percentage of binge drinking population), P_HEALTHY (percentage of healthy population).

GeoDa -> Explore -> Histogram
-> Boxplot
-> Scatterplot
-> Bubble Chart

Practice following statistical functionalities with variables of 'OBESE' (obesity population) and 'Population' (residential population)

GeoDa -> Map -> Rates-Calculated map -> Raw Rate
-> Excess Risk

With above functions, risk of obesity will be spatially illustrated in London, which can support further risk analysis.

Problem solving: Initially explore spatial accessibility in London

Functionality: Basic spatial analysis tools in QGIS

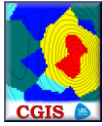
Data set: Maps of London Boroughs, Motorway and Underground Stations

In QGIS

Add "london_boroughs.shp", "motorway.shp" and "UndergroundStations.shp"

Practice following statistical functionalities:

QGIS -> Vector -> Analysis Tools -> Sum line lengths
-> Count points in polygon



With above functions, it is easy to find out figures related to local spatial accessibility, such as length of Motorway and number of tube stations.

Task 2: Thematic Mapping of Crime Data across London Boroughs

Problem solving: Create thematic maps to show spatial pattern of crimes in London

Functionality: Mapping functions in GeoDa

Data set: Crime variables by London Borough

In GeoDa

On "london_life_polygon.shp", practice following mapping functionalities with crime variables of "BURGLARY" (counts of burglary per 1000 persons), "CRIME_DANAG" (counts of crime damage per 1000 persons), "DRUG_OFFEN" (counts of drug offence per 1000 persons), "VEHICLES" (counts of vehicles theft per 1000 persons), "VIOLENCE" (counts of violence crime per 1000 persons).

GeoDa -> Map -> Quantile Map

-> Percentile Map

-> Box

-> Standard Deviation Map

-> Cartogram Map

With above functions, various thematic map can be created to demonstrate spatial distribution of different types of crimes

In QGIS

On "london_life_polygon.shp", practice following mapping functionalities with crime variables of "CRIME_1000" (total counts of crime per 1000 persons) with your preferred colour schemes:

QGIS -> Layer -> Layer Properties -> Symbolology

(right click of mouse -> Properties -> Symbolology

-> Graduated)

QGIS and GeoDa have similar functions on thematic mapping. QGIS is more flexible while GeoDa is more straightforward.

Please export rate and risk maps of obesity created in GeoDa from **Task 1**, export thematic map of crime created in QGIS from **Task 2** and put them into your portfolio.

GeoDa: right click of you mouse -> Save Image as

QGIS: Project -> Import/Export -> Export Map to Image

Please have comments on difference between functionalities of Geostatiscs and corresponding functionalities in conventional statistical packages.

In thematic maps, is there any difference in spatial pattern by different classifications and colour schemes?